



Research article

Feasibility and optimal sizing analysis of hybrid PV/Wind powered seawater desalination system: A case study of four ports, Egypt

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ARTICLE INFO

Keywords:

Sea water desalination
Hybrid PV/Wind
Optimization
Homer
Economic evaluation

ABSTRACT

This research aims to investigate A novel and complete system consists of hybrid renewable energy farm with high-energy-consuming seawater desalination in fourth locations in Egypt. This paper proposes fuzzy-based multi-criteria decision-making model for optimal sizing of a hybrid PV/Wind/Storage system to power the reverse osmosis (RO) desalination process in order to increase freshwater availability and meet the electric load requirement in selected area. Firstly, according to collected data and cost of electricity (COE) for various renewable energy resources, specific procedures to determining the optimal system design and optimal sizing (solar cell, wind turbine parameter and storage system units) are presented for selected site. The optimization based on finding the minimum net present cost (NPC) feeds the load demand and maintains the system's reliability. Secondly, the fuzzy analytic hierarchy process (FAHP) is implemented to select the optimal system considering ten performance criteria. The results show that the feasible design consists of 16×100 -kWwind turbines, 4127-kW photovoltaic array, 153 PH245 storages and 710-kW converter. The feasible design has the optimal economic values among all criteria with least NPC, COE, and payback-period (PBP) of \$9,08,046, 0.091\$/kWh and 1.1 yr, respectively. Besides, it has a 100 % renewable energy system (RES).

In contrast to recent studies that concentrated on incorporating renewable energy sources into desalination systems at specific Egyptian location. This approach is applied to different case studies with diverse renewable energy resources in multiple locations across Egypt, thus providing valuable insights for decision-makers tackling electricity and water shortages throughout Egypt. This research fills a gap in the optimization of hybrid renewable energy systems (HRES) by taking into account a variety of sustainability criteria, including technological feasibility, cost-effectiveness, emissions reduction, and socio-political considerations, in contrast to other studies that concentrated on single aims. Additionally, the research presents fuzzy logic and fuzzy-AHP and fuzzy-VIKOR decision-making techniques, enabling a comprehensive assessment of system design alternatives while taking uncertainties into consideration. In general, this study makes a significant addition to the field of renewable energy integration and gives decision-makers a framework for choosing the best answers to Egypt's water and electricity problems.

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<https://doi.org/10.1016/j.heliyon.2024.e40313>

Received 1 January 2024; Received in revised form 24 October 2024; Accepted 8 November 2024

Available online 10 November 2024

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List of acronyms

RO	reverse osmosis
FAHP	Fuzzy analytic hierarchy process
RESs	renewable energy systems
PV	Photovoltaic
STC	Solar Thermal Collectors
NPC	net present cost
COE	cost of electricity
HOREF	hybrid offshore renewable energy farm
AHP	analytic hierarchy process
MCDM	Multi-Criteria Decision Analysis
KPC	key performance criteria
PBP	payback-period
HRES	Hybrid renewable energy system
DPs	desalination plants
DS	desalination system
DGen	Diesel generator
GHG	greenhouse gas
UN	United nations
CSR	capacity shortage ratio
CEI	carbon emission impact
SP	Surplus power
LR	Land requirement

1. Introduction

One of the principal tributaries of the Nile River, the Blue Nile, is the site of the dam construction. The downstream flow of the Nile River may be impacted when the dam is constructed and the reservoir behind it fills up. Egypt's hydropower production and access to water resources may be impacted by this change in river flow [1].

Egyptian remote areas suffer from electrical energy and freshwater shortages. In these remote areas, the freshwater demand is satisfied through water tanks, at very high costs, that could even over 10€/m³ [2]. To overcome poverty and secure that all countries will enjoy prosperity by 2030 UN1 observed the Sustainable Development Goals, Clean energy becomes an important goal for all countries. So, the transformation to harvest energy from (RESs) has become an imperative specially with the increasing of energy demands and clean water shortages [3].

In the last years water desalination technology being one of the most applicable solutions for freshwater production. Two basic process groups may be distinguished among the various extant desalination technologies: procedures based on evaporation (phase change) and processes based on membranes (non-phase change) [4]. Desalination operations are frequently powered by fossil fuels, which have a large environmental impact. Furthermore, fossil fuels are often rare in isolated populations such as islands or coastal locations. As a result, the prospect of using renewable energy sources to power desalination processes is the optimal solution [5]. The process of desalinating saltwater using solar energy is known as "solar desalination" in a solar desalination plant, the process is driven by electricity produced by Photovoltaic (PV) panels or heat supplied by Solar Thermal Collectors (STC) [6].

Egypt has been implementing reverse osmosis desalination projects to address the growing demand for freshwater. Here are some notable projects:

- 1 West Port Said Desalination Plant: This project is part of the government's plan to provide clean drinking water to all citizens by 2030. The plant has a capacity of 150,000 cubic meters per day and uses reverse osmosis technology to treat seawater.
- 2 El Galala Desalination Plant: Located in the Red Sea governorate, the El Galala plant has a capacity of 20,000 cubic meters per day and uses reverse osmosis technology to treat seawater.
- 3 Ain Sokhna Desalination Plant: This project is a joint venture between the government and the private sector. The plant has a capacity of 150,000 cubic meters per day and uses reverse osmosis technology to treat seawater.
- 4 Marsa Alam Desalination Plant: Located in the Marsa Alam region, this plant has a capacity of 20,000 cubic meters per day and uses reverse osmosis technology to treat seawater.
- 5 Borg El Arab Desalination Plant: This project is part of the government's plan to provide clean drinking water to the growing population in Alexandria. The plant has a capacity of 24,000 cubic meters per day and uses reverse osmosis technology to treat seawater.

To reduce greenhouse gas emissions and mitigate the impacts of climate change, Egypt has been investing in renewable energy projects to power its desalination plants, with a focus on reverse osmosis (RO) technology. Here are some examples of RO desalination projects in Egypt that are powered by renewable energy.

- 1 Al Khafji Solar Desalination Plant: This is one of the largest solar-powered desalination plants in the world, located in the city of Al Khafji. The plant uses both photovoltaic (PV) and concentrated solar power (CSP) technologies to generate electricity for the RO

desalination process. It has a capacity of 60,000 cubic meters per day and provides water for domestic, commercial, and industrial use.

- 2 Marsa Matrouh Solar Desalination Plant: This plant, located in the city of Marsa Matrouh, uses PV technology to generate electricity for the RO desalination process. It has a capacity of 24,000 cubic meters per day.
- 3 Abu Dhabi Fund for Development Solar Desalination Plant: This plant, located in the city of Shalateen, uses PV technology to generate electricity for the RO desalination process. It has a capacity of 500 cubic meters per day.
- 4 Ras Sudr Solar Desalination Plant: This plant, located in the city of Ras Sudr, uses PV technology to generate electricity for the RO desalination process. It has a capacity of 500 cubic meters per day.
- 5 Aswan Solar Desalination Plant: This plant, located in the city of Aswan, uses PV technology to generate electricity for the RO desalination process. It has a capacity of 500 cubic meters per day.

While solar desalination plants have several advantages, such as using a renewable source of energy and producing no greenhouse gas emissions, they also face some challenges. Here are some of the main problems faced by solar desalination plants [7].

- 1 High capital costs: Solar desalination plants require a significant investment in solar panels, storage batteries, and desalination equipment, making them more expensive to build than conventional desalination plants.
- 2 Limited output: Solar desalination plants are dependent on the availability of sunlight, which means their output can be limited by cloudy weather, seasonal variations, or shorter days during the winter months.
- 1 High maintenance costs: Solar panels require regular maintenance and cleaning to ensure they are functioning efficiently, which can be challenging in desert environments where dust, sand, and other debris can accumulate quickly.
- 2 High energy consumption: Desalination plants, including solar desalination plants, require a significant amount of energy to operate, which can be difficult to generate from solar power alone. This means that backup energy sources, such as batteries or generators, may be necessary to ensure a reliable supply of fresh water.
- 3 Scaling up: While solar desalination plants have been successfully implemented on a small scale, scaling up these plants to meet the water needs of larger populations can be challenging due to the high capital and maintenance costs involved.

Hybrid PV/Wind/Storage system was implemented to power the RO desalination process in order to increase freshwater availability and meet the electric load requirement of a stand-alone region in isolated areas in Egypt. Hybrid renewable energy systems (HRES) are an optimal solution because they can provide a feasible cost and performance improvements, and they can be adjusted to meet a load profile requirement, and they might be on grid or off grid [8].

1.1. Design and implementation of renewable energy systems (RES) desalination projects

Some factors should be taken into account to design the desalination project coupled to RES technologies as flow [9–11]:

- Salt water source and type.
- Fresh water amount demand.
- Distance from electrical grid.
- Land and site characteristics.
- Economic situation.
- Renewable energy resources.
- Environmental aspects.
- Standard living.

These factors are applied to identifying the best desalination technology for a remote areas location. The optimal system should be the most feasible model. It should supply acceptable amounts of freshwater for the duration of the project. Isolated areas suffer from electricity shortages thus makes the off grid systems is the suitable solution for these areas. To achieve the fresh water needs in remote areas, hybrid offshore PV/wind desalination system is installed [12]. In locations where connected to electrical grid on grid application is the most optimal solution. In on grid system excess electricity can be sold back to the electrical network. Another important factor is the control topology of the system. Renewable energy resources (wind speed & solar radiation) are important factors considerations in remote areas applications. The optimal capacity calculated by the energies available, water storage capacity, and the freshwater demand. But, the final decision will be made based on the available budget [13,14].

1.2. Research motivation and main goal

When the annual water content per capita is reduced by 665 m3, the reliability of agriculture projects and the well-being of human beings are put at risk. Despite the commendable efforts made by the Egyptian government to address the electrical sector's challenges and demands, the impact of water scarcity remains a significant concern for sustainable development. More than 250 villages in Egypt still lack access to clean water and energy, despite the government's admirable efforts to alleviate bottlenecks in the electrical sector. Egypt's government implement long-term strategy to achieve sustainability objectives and targets across the board called Egyptian Vision 2030 A portion of the goal is to increase the amount of freshwater by building more than fourty desalination plants (DPs) by

2025 and to increase the share of renewable energy to 49.5 GW in 2029/30 and 62.6 GW in 2034/35, respectively. One efficient way to accomplish these goals is to integrate (HRESs) with DPs fed from either brackish or ocean. Thus, the main objective of this work is to develop an effective model for assessing and enhancing the design and viability of an integrated large-scale RO fed by HRESs that integrates five scenarios and considers various design factors.

The increasing global demand for potable water, driven by population growth and urbanization, has led to the need for sustainable and environmentally friendly desalination technologies. Reverse osmosis (RO) desalination is one such technology, and integrating it with hybrid renewable energy sources can significantly reduce its environmental impact and improve its long-term viability. The motivation for the optimal design of hybrid renewable energy-based RO desalination systems can be broken down into several key factors.

1. Water Scarcity and Growing Demand

Water scarcity is a pressing issue faced by millions of people worldwide, particularly in arid and semi-arid regions. The growing demand for potable water necessitates the development of efficient and sustainable desalination technologies to ensure access to safe drinking water for all.

2. Environmental Impact and Climate Change

Traditional desalination methods, such as thermal processes, can have significant environmental impacts due to their high energy consumption and greenhouse gas emissions. By using renewable energy sources, the carbon footprint of desalination can be significantly reduced, helping to mitigate the effects of climate change.

3. Energy Security and Reliability

Integrating renewable energy sources into desalination systems can reduce reliance on fossil fuels and improve energy security. Furthermore, hybrid renewable energy systems can provide greater reliability, as they can continue to function even if one energy source is temporarily unavailable.

4. Technological Advancements

Recent advancements in renewable energy technologies and RO desalination have made it more feasible to combine these systems. Improved efficiency, reduced costs, and increased durability of renewable energy systems make them more attractive for integration with desalination plants.

5. Cost Optimization

A critical aspect of designing hybrid renewable energy-based RO desalination systems is cost optimization. By selecting the appropriate combination of renewable energy sources and optimizing system design, it is possible to reduce both capital and operational costs, making the system more affordable and sustainable in the long run.

6. Social and Economic Benefits

Implementing hybrid renewable energy-based RO desalination systems can lead to job creation in the renewable energy and desalination sectors, boosting local economies. Additionally, access to clean drinking water can improve public health and overall quality of life, particularly in regions facing water scarcity.

In summary, the motivation for the optimal design of hybrid renewable energy-based RO desalination systems lies in the need to address water scarcity, minimize environmental impact, improve energy security and reliability, capitalize on technological advancements, optimize costs, and promote social and economic development. By achieving these goals, such systems can play a significant role in ensuring sustainable access to potable water for communities worldwide.

1.3. Literature review

Various papers have investigated the HRES supply various load profile with Several configurations in different locations and observed their technical and economic analysis. Usually, the optimal design characterized by lowering their COE or NPC.

In [15] the author implements an economic study of a hybrid PV/wind/battery energy-driven hydrogen generation system in the Ras Ghareb Region of Egypt. An examination of Egypt's green hydrogen production using a hybrid energy system is conducted. Priced at \$2.22 per kilogram. In Ref. [16] Author execute three 100 % renewable energy systems for Estonia. Hydrogen fuel production and district heating are two uses for surplus energy. The ideal 194 GW hybrid PV/wind system is primarily wind-powered. The hybrid system that works best generates 15.05×10^9 tons of hydrogen fuel.

In [17] In the Rajshahi region, the study looks into grid-connected and standalone hybrid PV/Diesel/Batt systems with a focus on residential load profiles. The COE of the implemented system is 0.28\$/kWh. In Ref. [18] The authors apply techno-economic analysis

and optimal sizing for hybrid PV-battery-cooling storage systems in commercial buildings in China.

In [19] authors implemented Economic Analysis of Off-Grid Solar PV Desalination: An off-grid solar PV system for small-scale desalination units was the subject of an investigation. The goal of the research was to use renewable energy sources, such as solar power, to reduce water consumption in an environmentally beneficial manner. According to the study's findings, off-grid solar PV systems are an economical and sustainable way to desalinate water. In Ref. [20] The study examined the financial implications of utilizing waste heat recovery from wind turbine generators for seawater desalination through the humidification-dehumidification (HDH) method. The study demonstrated the possible financial advantages of waste heat recovery for desalination by contrasting the performance and cost characteristics of a hybrid HDH-RO unit powered by wind turbines with a solo RO unit.

In [21] researchers used solar and wind power for desalination to meet rising global freshwater demand Integrate renewables with tradition for reliable, cost-effective energy. The study centers on a desalination plant in Kerkennah, Sfax, Tunisia. In Ref. [22] study describes the best possible integration of membrane-based saltwater desalination units with wind turbines and/or photovoltaics in a hybrid renewable energy system to meet freshwater demands in Rio Grande and Camarones, two cities in the southern region of Argentina. In both locations, freshwater prices from hybrid solar/wind systems range from 0.60 to 0.66 dollars per cubic meter.

In [23] The authors take into account the intermittent nature of renewable energy sources and provide a scaling approach that takes battery storage, load demand, and renewable energy supplies into account. Techno-economic analysis and optimization algorithms aid in evaluating system performance and determining the design that is most economical. In Ref. [24] In the paper, a genetic algorithm-based optimization methodology for a hybrid PV/wind/battery system in seawater desalination is introduced. The goal of the study is to reduce overall costs, which include startup, ongoing, and maintenance expenses. The authors optimize system efficiency and cost-effectiveness by determining the best configuration and sizing of each component by using the genetic algorithm.

In [25] The most economical plan's calculated COE was \$0.1074/kWh, and the high percentage of renewable energy effectively cut emissions by 94 %. In contrast to typical HRSES applications, the viability and optimization of an energy-water-heat nexus supporting the airport facilities in Egypt.

In [26] researchers investigated a hybrid PV/wind and diesel engine desalination system (DS) for RO water treatment. Authors observed that a hybrid PV/wind fed RO plant is feasible solution for remote areas. In Ref. [27] authors implemented optimal sizing for various designs of PV fed DS and selected the feasible system. The research showed that PV fed RO technology is the most feasible and optimal DS to supply freshwater to remote areas in Saudi Arabia. In Ref. [28] presented the projects of RES-fed Desalination unit in Greece. In [29], authors implemented an optimal sizing for a solar fed RO using the response surface approach. The system was implemented and designed for the desalination of brackish water. In Ref. [30] authors proposed RESs supply a DS used for brackish water and observed that there is around 50 % reduction in mean water permeability coefficient while the specific electrical energy consumption was raised from 1.82 kWh//m³ to 2.21 kWh/m³. In Ref. [31], authors observed that the intermittency nature of RESs didn't have negative effect on the behavior of the membrane in short-term periods. In Ref. [32] researchers observed that ROs, fed by RESs, can work under fixed or variable conditions. In Ref. [33] authors showed that some scenarios, for lower renewable energy resources such as lower solar irradiance, the amount and quality of the produced water can be reduced. In Ref. [34] the researcher presented different desalination technologies, to supply fresh-water, which are fed by HRESs with integrated storages, and diesel generator (DGen) to improve system reliability. In Ref. [35] author presented another economic evaluation system of using HRESs s for small DPs depend on its life cycle cost. The research observed that the PV/battery storage design is the most feasible HRESs for RO because the low oil cost and high solar radiation in Oman.

In [36] author conducted a techno-economic analysis of a hybrid solar-wind-diesel-powered RO system in the city of Marsa Alam, Egypt. The study found that the optimal design involved a 100 kW wind turbine, a 50 kW photovoltaic array, and a 100 kW diesel generator, with a total capacity of 150 m³/day. The study also highlighted the importance of optimizing the operation of the system to minimize fuel consumption and maximize the utilization of renewable energy sources.

Table 1
A synopsis of the latest research on RESs based on DPs.

Reference year	Research type	Capacity M ³ /day	Electrical demand kWh/day	location	System design	Sensitivity analysis	Research object
[36] 2019	Theoretical	150	755	Egypt	Pv + wind + Dgen	No	Feasibility & Inviroment
[37] 2020	Theoretical	2.5	11	Egypt	PV + wind	Yes	Feasibility
[38] 2017	Theoretical	10	55	Egypt	PV + wind	Yes	Feasibility
[40] 2005	Theoretical	300	1525	Lybia	Pv + wind + Dgen	NO	Feasibility
[41] 2022	Theoretical	10,000	59140	Jordan	Pv + wind + Dgen + storage	NO	Technical & economical
[42] 2019	Theoretical	100	500	gran canaria	PV + BS	Yes	Technical & economical
[43] 2021	Theoretical	1000	2400	—	PV + BS	NO	Feasibility Inviroment
[44] 2022	Theoretical & practical	50000–190000	250000–950000	Egypt	PV + BS	yes	Technical
[45] 2023	Theoretical	5	22	Indonisia	PV + BS	NO	Feasibility & Inviroment

In [37] researcher proposed a hybrid renewable energy-based RO system for a remote village in Egypt that is not connected to the grid. The system combines solar and wind energy sources, with a 5 kW wind turbine and a 2.5 kW photovoltaic array, to power a 2.5 m³/day RO unit. The study found that the proposed system could provide a reliable and cost-effective solution for providing fresh water to the village.

In [38] author conducted a techno-economic analysis of a hybrid renewable energy-based RO system for a remote island in Egypt. The system combines solar and wind energy sources, with a 12 kW wind turbine and a 20 kW photovoltaic array, to power a 10 m³/day RO unit. The literature showed that the hybrid system provides a reliable and cost-effective solution for meeting the remote areas water demand. Recent case-researches on RODPs fed from HRESs is shown in Table 1. Since the economic performance of HRSEs is influenced by a number of interrelated factors as well as their design, multi criteria decision making (MCDM) analysis is used to evaluate the predetermined options in order to select the most suitable system by evaluating them based on a wide range of criteria. Generally speaking, there are two types of MCDM methods: classic and fuzzy-based [39].

1.4. Literature gaps and study contributions

Upon deeper inspection, several observational and research gaps are identified in the literature, including.

1. **Limited Multi-Location Studies:** Most of the study that has been done so far has been on certain areas. There is a gap in comprehensive studies that consider multiple locations to assess the feasibility and optimal sizing of hybrid PV/Wind powered seawater desalination systems.
2. **Limited comparison Analysis:** To identify the most dependable and affordable seawater desalination options, there aren't enough comparison studies in the literature comparing various system configurations, such as PV/WT with and without backup diesel generators. Studying comparisons can give important information about the trade-offs between various system designs.
3. **Insufficient Techno-Economic-Environmental Integration:** Although some studies address techno-economic issues, there is a deficiency in studies that thoroughly integrate environmental and techno-economic factors for hybrid photovoltaic/wind powered seawater desalination systems. Making well-informed decisions requires a comprehensive strategy that takes into account system sustainability, environmental effect, and economic optimization.
4. **Irregular Fuzzy Logic Decision-Making** Prior research has mostly relied on HOMER software and other economic optimization tools, without including fuzzy logic decision-making models. The evaluation of system configurations based on several performance criteria is limited by the lack of fuzzy-based multicriteria decision-making methodologies.
5. **The lack of sufficient real-world data** on the performance of hybrid renewable energy-based reverse osmosis (RO) systems in Egypt is primarily due to the heavy reliance on simulation models and assumptions in existing studies. This is particularly evident in remote areas where grid electricity is not readily available.
6. **Consequently, the literature has given limited attention** to the inclusion of energy storage systems, which are vital for ensuring the reliability and flexibility of hybrid renewable energy-based RO systems. Further research is necessary to optimize the sizing and operation of energy storage systems, aiming to minimize costs and maximize system performance. By addressing these gaps, valuable insights can be gained to enhance the understanding and practical implementation of hybrid renewable energy-based RO systems in Egypt and similar contexts.
7. **Neglect of the socio-economic impacts:** The existing literature often overlooks the significance of considering the socio-economic effects associated with hybrid renewable energy-based reverse osmosis (RO) systems. It is crucial to conduct comprehensive techno-economic analyses to evaluate the social and economic advantages of these systems, particularly in isolated and underprivileged areas. Further research is needed to bridge this gap and provide a more holistic understanding of the socio-economic impacts of hybrid renewable energy-based RO systems.

Although Egypt has conducted studies on the optimal design of hybrid renewable energy-based reverse osmosis (RO) systems, there are still significant gaps in the existing literature that require further exploration. Some of these gaps include.

1.5. Contribution:

Hybrid renewable energy-based reverse osmosis (RO) systems hold significant potential in meeting Egypt's water demand. However, there are still several gaps in the existing literature that need to be addressed. To enhance the understanding and optimize the performance of these systems, further research is necessary in the following areas:

Consider multiple locations to assess the feasibility and optimal sizing of hybrid PV/Wind powered seawater desalination systems. This research takes into account variations in solar irradiation, wind patterns, and water availability across different locations. By evaluating multiple sites, we can provide more robust insights into the performance and potential of these systems.

Comparative Analysis: More comparison studies should be conducted to evaluate and compare different system configurations for seawater desalination. This includes comparing PV/WT systems with and without backup diesel generators. Such analyses can provide valuable information on the trade-offs between system designs, considering factors such as reliability, cost-effectiveness, and environmental impact.

Develop fuzzy logic decision-making models to evaluate system configurations based on multiple performance criteria. By incorporating AHP MCDM methodologies, the evaluation of hybrid PV/Wind powered seawater desalination systems can consider various factors simultaneously, energy production, cost, reliability, environmental impact, and social acceptance. This lead to more

comprehensive and informed decision-making processes.

Gathering actual data: There is a need to collect real-world data on the performance and operation of hybrid renewable energy-based RO systems in Egypt. This will provide valuable insights into system efficiency, reliability, and overall effectiveness, helping to validate and refine existing models and assumptions.

Improving energy storage systems: Energy storage plays a crucial role in ensuring the dependability and flexibility of hybrid renewable energy-based RO systems. More research is required to optimize the sizing, design, and operation of energy storage systems. This will help minimize costs, improve system stability, and enhance overall performance.

Analyzing socioeconomic effects: The socioeconomic impacts of hybrid renewable energy-based RO systems need to be thoroughly evaluated. This includes conducting comprehensive techno-economic analyses to assess the social and economic advantages of these systems, particularly in isolated and underprivileged areas. Such studies will provide valuable insights for decision-makers and stakeholders.

Appraising environmental effects: While hybrid renewable energy-based RO systems are considered environmentally friendly, a comprehensive analysis of their environmental effects is still needed. This involves evaluating water usage, greenhouse gas emissions, and other potential environmental impacts. By conducting thorough investigations, the design and operation of these systems can be tailored to minimize their environmental footprint.

Addressing these gaps in the literature will contribute to the improved performance and sustainability of hybrid renewable energy-based RO systems in Egypt. By gathering actual data, enhancing energy storage systems, analyzing socioeconomic impacts, and appraising environmental effects, a more comprehensive understanding can be achieved, leading to more efficient and effective implementation of these systems.

1.6. Research structure

To give an overview of the research structure, part 2 discusses the methodology and optimization model's research approach. Part 3 shows the investigation site's characteristics, load profile and system description. Section 4 discusses the generated HRES-RO components costs. Section 5 provides the results and discussion. Conclusions and of the study are condensed in part 5.

2. Methodology

The primary main goal of this study is to identify the most suitable alternative plan for (HRESs) by exploring various configurations involving PV, Wind, diesel and storage units. These HRESs are specifically designed to meet the electricity requirements of a large-scale Reverse Osmosis (RO) plant located in selected areas.

Hence, an approach of feasibility and optimization is implemented to find the feasible HRES alternatives using HOMER software). The performance evaluation of each option considers several key factors, including (NPC), (COE), capacity shortage ratio (CSR), and carbon emission impact (CEI). Design constraints load balance and storage safety limits are also taken into account. To achieve

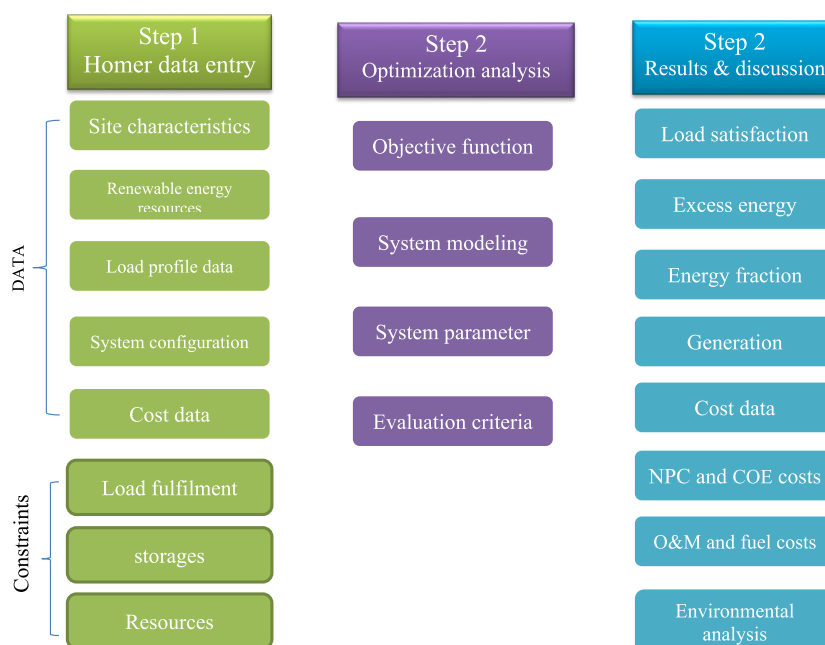


Fig. 1. Presented description of the research.

comprehensive multi-criteria optimization, a combination of (AHP) and (MCDM) methods is implemented. The AHP is employed to assign weights to the different (KPC), and subsequently used for the final ranking of alternatives.

Fig. 1 provides an overview of the research methodology employed in this study, illustrating the general flow of the process. Further descriptive details of the techniques utilized are presented in the subsequent subsections.

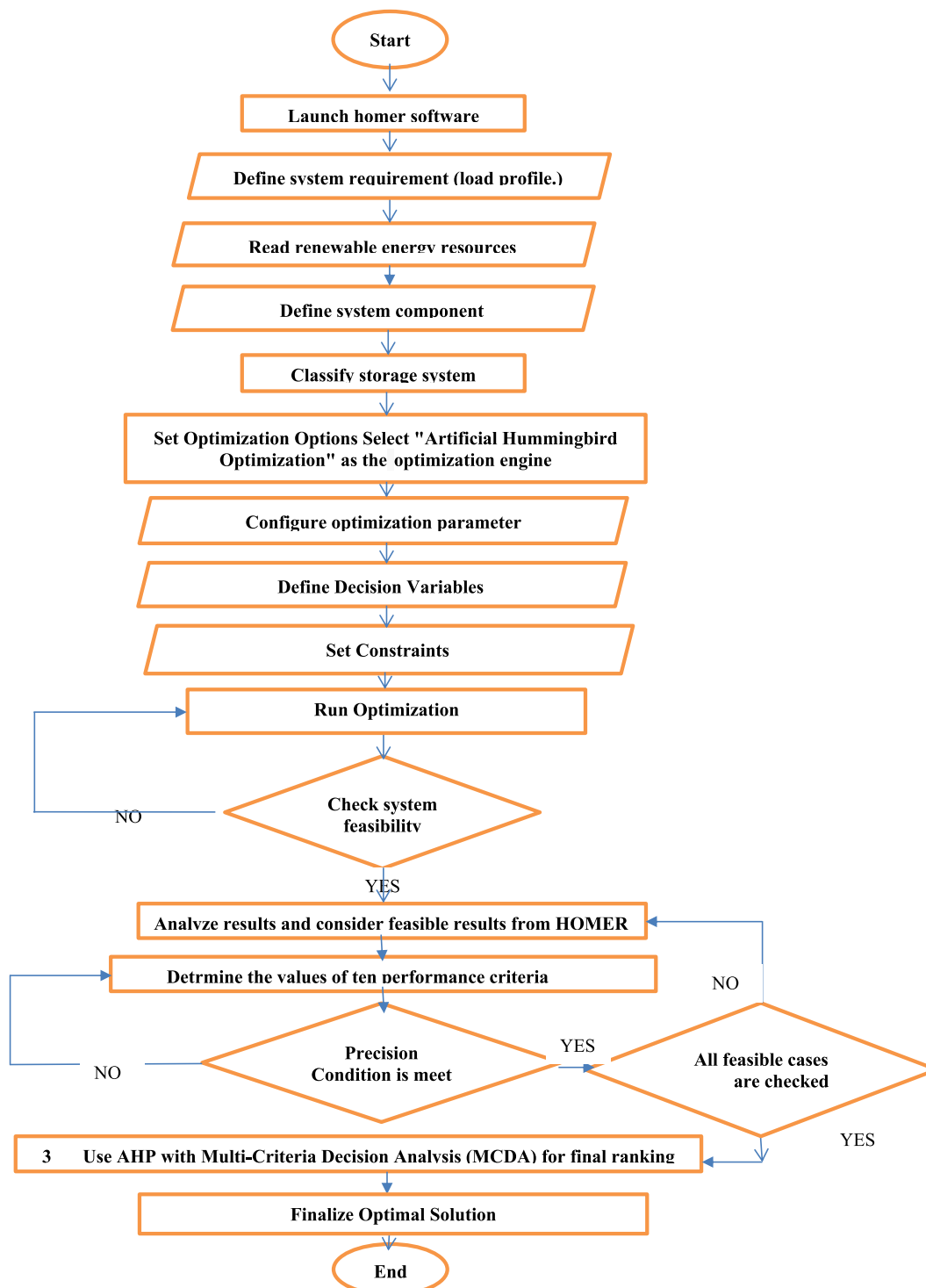


Fig. 2. Flowchart of the proposed model.

1. Define Requirements:
 - Determine the required daily water production (in cubic meters or gallons).
 - Specify the desired energy source (renewable energy).
 - Identify any specific constraints or limitations (e.g., available space, budget).
2. Assess Renewable Energy Availability:
 - Evaluate the available renewable energy sources in the location (e.g., solar, wind, tidal).
 - Determine the average energy production potential based on historical data or feasibility studies.
3. Calculate Energy Demand:
 - Estimate the energy requirements for the desalination process, including pre-treatment, pumping, and membrane operation.
 - Consider any energy losses during conversion or transmission.
4. Run HOMER Software:
 - Input the system requirements, including water demand, energy demand, renewable energy sources, and system constraints into the HOMER software.
5. - Review the results and make necessary adjustments if required.

2.1. Optimization using HOMER

The HOMER is an investigational tool that is used to help in optimization process. HOMER has been used in several energy research, as reported in Ref. [46], because to its dependability, unified programming design, and friendly user-interface. Economic information, such as the project lifetime and real interest rate, is crucial for the Homer simulation. It enables the conversion of one-time expenses into annualized expenditures. The Homer analysis incorporates capital, replacement, and operation and maintenance costs [47]. The (NPC), which represents the total cost of installing and maintaining the system over its lifespan, serves as the foundation for the Homer optimization computation. In addition to that the COE is the average cost per kWh (\$/kWh) of the electrical energy produced by the system as a result, the optimization in this paper, has three steps of implementation, as shown in Fig. 2.

First, load demand, component costs, and climatic data are assessed as part of a main evaluation based on the proposed approach.

The optimization core is the second layer, which models and simulates preset energy technologies while taking into account a variety of limitations. HOMER begins work on a variety of HRESs options.

Finally, in the third layer, each viable case is assessed based on its feasibility, reliability, and environmental characteristics, using CSR, PbP, NPC, COE, SP, and CEI evaluation criteria.

The Focus Factor's default setting is 50, which generates quick results and is ideal for developing and iterating. Before finalizing a design, we run the optimization again with a lower Focus Factor (e.g., 5 or 10) to be sure that the selected solution is the best. When you press the Calculate icon, HOMER executes a number of optimizations: one for each Search Space combination, as well as one for each system category.

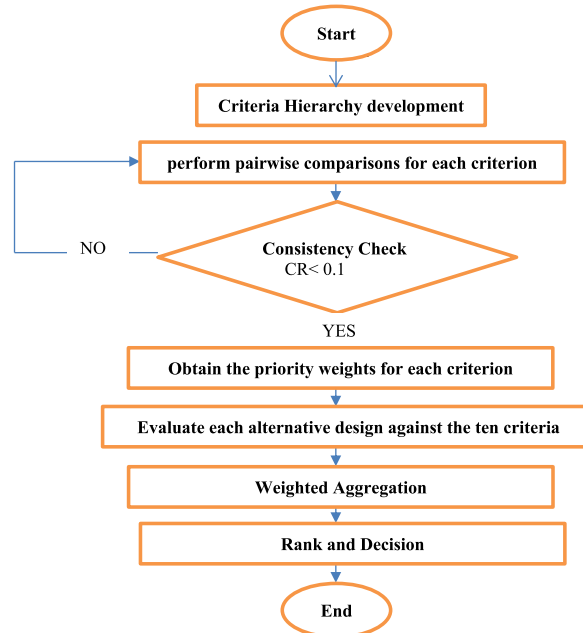


Fig. 3. Flowchart of the-AHP with MCDA method.

2.2. AHP with MCDA to choose the best design for a hybrid renewable energy system

For the current study, a novel AHP with MCDA method approaches was implemented for selecting the single optimal HRESs considering ten (KPC).

1. (NPC): Represents the total cost of the system, including initial investment and future cash flows, discounted to present value.
2. (COE): Indicates the cost per unit of energy generated by the system.
3. Operating and Maintenance (O&M): Reflects the ongoing costs associated with operating and maintaining the system over its lifetime.
4. Carbon Emissions (CE): Measures the amount of greenhouse gas emissions produced by the system during operation.
5. Renewable Energy Fraction (REF): Indicates the proportion of energy generated from renewable sources in the system.
6. Land Requirement (LR): Represents the amount of land area needed for installing and operating the system.
7. Payback Period (PP): Represents the time required for the system to recover its initial investment through energy savings or revenue generation.
8. (CSR): Reflects the system's ability to meet the energy demand without any capacity shortage.
9. Surplus Power (SP): Indicates the excess power generated by the system beyond the energy demand.
10. Installation Easiness (IE): Represents the ease of installing and setting up the system.

Here's how you can apply AHP with MCDA to choose the best design as show in Fig. 3.

- **Criteria Hierarchy:** Create a hierarchical structure with the main objective (e.g., "Select the best design for a hybrid renewable energy system") at the top, followed by the ten performance criteria listed above.
- **Pairwise Comparisons:** perform pairwise comparisons for each criterion. For example, compare NPC with COE and assign a value representing the relative importance. Repeat this process for all possible pairs of criteria.
- **Consistency Check:** Calculate the consistency ratio to ensure the consistency of judgments. Adjust the pairwise comparisons if needed to achieve a satisfactory consistency ratio.
- **Criteria Weights:** Obtain the priority weights for each criterion based on the pairwise comparisons using the AHP method. These weights represent the relative importance of the criteria.
- **Alternative Evaluation:** Evaluate each alternative design against the ten criteria. Assign scores or values for each criterion based on the performance of the design. For example, for NPC, a lower cost receives a higher score.
- **Weighted Aggregation:** Multiply the scores of each alternative by the corresponding criteria weights obtained from AHP. Sum up the weighted scores for each alternative to obtain an overall performance value.
- **Rank and Decision:** Rank the alternatives based on their overall performance values. The design with the highest value is considered the best choice for the hybrid renewable energy system.

2.3. Why AHP with MCDA to choose the best design for a hybrid renewable energy system

- **Appropriateness for Decision-Making:** The methodical organization of intricate decision problems is made possible by the widely-used Analytic Hierarchy Process (AHP). It helps to rank criteria and options according to their relative performance and importance, and to make well-informed decisions.
- **Handling Multiple Criteria:** When dealing with multiple criteria that must be taken into consideration at the same time, AHP is especially helpful. The study uses a variety of parameters to evaluate the effectiveness and practicality of various system

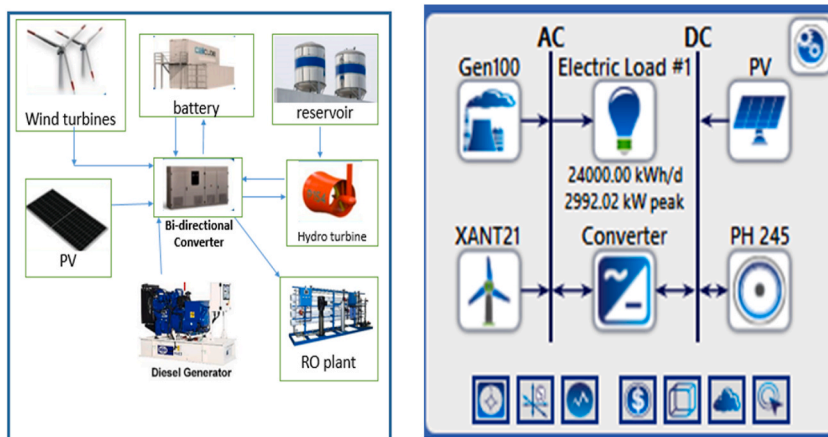


Fig. 4. Proposed system layout.

configurations for hybrid PV/wind powered seawater desalination. AHP facilitates the handling of various factors and offers a methodical framework for making decisions.

- **Transparency and Reproducibility:** By dividing complicated issues into a hierarchical structure and performing pairwise comparisons, Analytic Hierarchy Process (AHP) offers a transparent and systematic framework for decision-making.
- **comprehensive Analysis:** Because AHP-MCDA can offer a thorough analysis that takes into account a variety of variables and stakeholder preferences, we selected it for prediction and optimization.

3. Proposed system description

The study focuses on analyzing various isolated areas in Egypt, considering different sites, using the HOMER software. Fig. 4 illustrates the application of the newly developed hybrid renewable energy system to supply power to large-scale Reverse Osmosis Desalination Systems (RODP) situated in fourth coastal regions. The primary purpose of installing these RODPs is to meet the fresh water demands of the selected sites. These locations have abundant salt water resources and renewable energy potential.

To enhance system reliability and decrease capital and maintenance expenses, fresh water storages are employed instead of batteries. This substitution is made due to the relatively short lifespan of battery systems. By utilizing fresh water storages, the system becomes more dependable and cost-effective in the long run.

3.1. Load profile

Fig. 5 Shows the load characteristics of a desalination unit for all location under this research. The rated capacity of the desalination plant is 15,000m³/day, and average load are 1833 kW. The RODP unit section consists of two racks (2 × 7500) m³/day. whose configuration is based on 1 step – 1 stage system, with a total of seven elements per pressure vessel.

3.2. Description of integrated renewable energy sources

For this research, fourth locations are chosen across different ports in Egypt, namely Alexandria, Alsallom, Suez and Shalaten. To estimate the monthly average solar global horizontal irradiance and monthly average wind speed at these locations, NASA surface meteorology data is utilized. This data is derived from satellite observations [48–50]. The latitude and longitude coordinates of the selected study areas are provided in Table 2.

3.2.1. Solar resources

The monthly average solar radiation is shown in Fig. 6. Table 3 presented the highest, lowest, and annual average solar radiation for each place [48].

3.2.2. Wind speed parameter

Fig. 7 shows the monthly average wind speed. Table 4 gives the highest, lowest, and annual average wind speed for each given place [51–53].

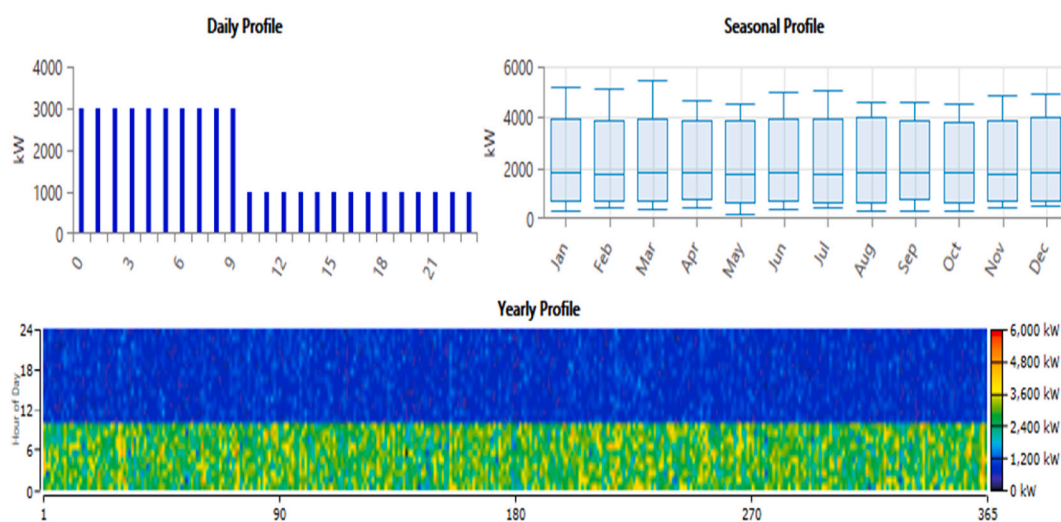


Fig. 5. Load profile.

Table 2
The geographical location of all sites.

Site	Latitude	Longitude
Alexandria	31.5	25.5
Alsalloum	31.5	29.5
Shalateen	23.5	35.5
Suez	29.5	32.5

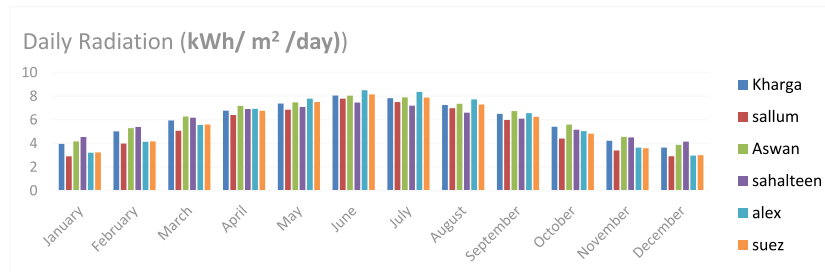


Fig. 6. Average monthly solar radiation.

Table 3
The highest, lowest, and annual average solar radiation for each place.

Site	Highest kWh/m ² /day	Month	Lowest kWh/m ² /day	Month	Average
Alexandria	8.5	June	2.97	December	5.87
Alsalloum	7.84		2.09		5.35
Shalateen	7.45		4.11		5.94
Suez	8.15		4		5.95

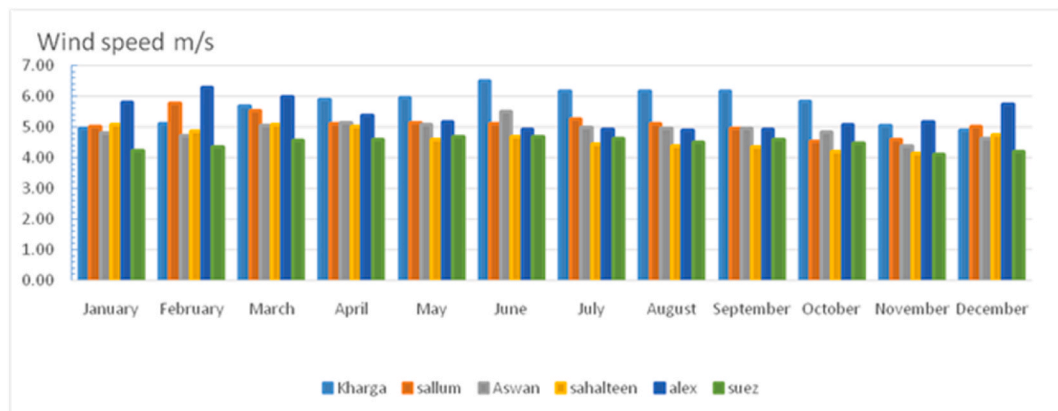


Fig. 7. Average monthly wind speed data.

Table 4
Highest, lowest, and annual average wind speeds for each place.

Site	Highest m/s	Month	Lowest m/s	Month	Annual average m/s
Alexandria	6.27	February	4.88	August	5.35
Alsalloum	5.77	February	5.54	October	5.09
Shalateen	5.08	January	4.14	November	4.62
Suez	6.8	May	5.2	November	5.5

4. System components costs and technical details

Table 5 and Table 6 provide an overview of the costs and technical specifications of the wind turbine, PV modules, and power converters. These details are inputted into the HOMER hybrid model to perform simulations for project lifetimes of 25 years.

5. Results and discussion

This section discusses the results along with a comprehensive analysis of the optimization results of the selected sites based on the (its economic, energy, and environmental performance). The most feasible subsystems are compared and assessed in accordance with the study's objectives and model constraints shown in Table 7. Tables 8–12 give the results of Five categories (PV standalone, WT standalone, hybrid system (PV and WIND), hybrid (WT and diesel), and hybrid (PV and diesel) have been implemented to organize the investigated configurations.

5.1. Finally, a comparison between the optimal systems is implemented to emphasize the implications of the proposed optimal case

5.1.1. Case 1 (PV standalone with PH245 storage)

The optimization results indicate that for the Suez location, a PV system with a capacity of 8093 kW and 243 units of PH245 storage is required to meet the annual energy consumption of the RO desalination system. The estimated annual energy production for this configuration is 15,382,088 kWh, with an excess electricity generation of 4,843,318 kWh. The calculated (NPC) for this configuration is \$14,100,765, and the (COE) is \$0.123/kWh. These results demonstrate that the Suez site requires the lowest number of PV panels among the selected sites due to its unique location, which benefits from favorable solar radiation conditions.

In contrast, for the Shalteen location, a PV system with a capacity of 14401 kW and 285 units of PH245 storage is needed to meet the annual energy consumption of the RO desalination system. The estimated annual energy production for this configuration is 28,491,432 kWh, with a significant excess electricity generation of 18,009,523 kWh. The NPC for this configuration is calculated to be \$22,300,725, and the COE is \$0.1970/kWh. These results indicate that the Shalteen site requires the highest number of PV panels among the selected sites due to its poor solar radiation conditions, which necessitate a larger PV capacity to meet the energy demand.

5.1.2. Case 2 (WT standalone with PH245 storage)

In case 2, 36 WT and 514 units of PH245 storage are required to meet the annual energy consumption of the RO system in Suez location. The estimated annual energy production for this configuration is 13,855,499 kWh, with an excess electricity generation of 3,798,802 kWh. The calculated (NPC) for this configuration is \$11,901,601, and the (COE) is \$0.1050/kWh. These results demonstrate that the Suez site requires the lowest number of WT panels among the selected sites due to its unique location, which benefits from favorable solar radiation conditions.

5.1.3. Case 3 (hybrid system PV and WT with PH245 storage)

In case 3, 16 WT, 4127 PV panels and 153 units of PH245 storage is required to meet the annual energy consumption of the RO system in Suez location. The estimated annual energy production for this configuration is 13,983,298 kWh, with an excess electricity generation of 4,068,772 kWh. The calculated (NPC) for this configuration is \$9,910,202, and the (COE) is \$0.0873/kWh. These results demonstrate that the hybrid system site is the most feasible system specially in in Suez location.

Table 5
Costs and technical details of system component.

PV		Power converters	
Model Name	Generic flat-plate PV	Rated power	1 kW
Tracking system	Fixed	Rectifier relative capacity	100 %
Ground reflectance	20 %	Inverter and rectifier efficiency	95 %
Rated power	1 Kw	Efficiency	90 %
Derating factor	0.5 %	Capital cost	600 \$
Operating temp	47 C	Replacement cost	560 \$
Efficiency	13 %	Lifetime	15 years
Capital cost	1200\$	Batteries	
Replacement cost	1000\$	Manufacturer	Gildemeister
Operation & Maintenance cost	25\$/year	Model	Cellcube®FB20-40
Lifetime	25 years	Nominal voltage	3.7 V
Storage Generic pumped hydro (PH245)		Nominal capacity	1 kWh
Manufacturer	Guinard energies	Maximum capacity	276 Ah
Model	Guinard- PH245	Capital cost	600 \$
Capacity	1000 m ³ of water	Replacement cost	550 \$
discharge period	12 h	Operation & Maintenance cost	10 \$/year
Diesel generator		Life time	10 years
Rated power:	100 kW		

Table 6
100 kW wind speed parameter.

	Characteristics	Specification
Main data	Model	EOX M-21
	Design class	IEC Class IA wind turbine
	Design life	30 years without major component replacement
	Rated power	100 kW
	Rated wind speed	10 m/s (36 km/h)
	Swept area:	347 m ²
	Specific area	3.47 m ² /kW
	Number of blades	3
Rotor	Power control	Stall
	Cut-in wind speed:	2,8 m/s
	Rated wind speed:	10 m/s
	Cut-off wind speed	20 m/s
Generator	Type	SYNC PM
	Voltage	400 v
Tower	Hub height	32 m

Table 7
Model constraints.

	Value	Value %
Daily water supply	15000 m ³ /day	
Wind speed	5 m/s	5.3 %
Solar radiation	5 kWh/m ² /day	8.2 %
CO2 emissions	0\$/kwh	0 %
Capacity shortage	0 %	03 %

Table 8
Cost Analysis of optimal systems (Net Present cost).

NPC	PV (Stand-alone)	WT(stand-alone)	Hybrid system (PV,WIND)	Hybrid (wind, diesel)	Hybrid (pv,diesel)
SUEZ	\$14.0	\$11.9	\$9.9	\$12.0	\$13.6
Alex	\$13.9	\$18.0	\$10.6	\$17.4	\$14.3
Sallum	\$19.3	\$11.9	\$12.0	\$12.0	\$19.3
Shalteen	\$22.3	\$19.3	\$11.4	\$17.8	\$22.3
Average	\$15.3	\$14.7	\$10.5	\$14.35	\$15.5

Table 9
Cost Analysis of optimal systems (cost of electricity).

COE	PV (Stand-alone)	WT(stand-alone)	Hybrid system (PV,WIND)	Hybrid (wind, diesel)	Hybrid (pv,diesel)
SUEZ	\$0.1230	\$0.1050	\$0.0873	\$0.106	\$0.120
Alex	\$0.1230	\$0.1600	\$0.0938	\$0.154	\$0.126
Sallum	\$0.1700	\$0.1050	\$0.1060	\$0.106	\$0.170
Shalteen	\$0.1970	\$0.1700	\$0.1010	\$0.157	\$0.197
Average	\$0.135	\$0.130	\$0.093	\$0.131	\$0.136

Table 10
Optimal results of HRES system.

System configuration	PV (kW)	No of wind turbine	Converter(kW)	No of PH245
Suez	4127	16	710	153
Alex	4170	19	735	179
Sallum	2983	38	510	184
Shalteen	3795	27	635	100

5.1.4. Case 4 (WT and Dgen with PH245 storage)

In case 4, 25 WT, 7 Dgen and 105 units of PH245 storage are required to meet the annual energy consumption of the RO system in Suez location. Which consider the most feasible system in this case with (NPC) \$12,201,301, and the (COE) is \$0.106/kWh with a CEI of 118,753.03 kg/yr and an RF of 91.12 %.

Table 11
Optimal results of PV system.

System configuration	PV (kW)	Converter(kW)	No of PH245
Suez	8093	1642	243
Alex	8211	1650	363
Sallum	11273	1950	352
Shalteen	14401	2450	285

Table 12
Optimal results of wind energy system.

System configuration	No of wind turbine	No of PH245
Suez	36	514
Alex	92	363
Sallum	36	514
Shalteen	86	491

5.1.5. Case 5 (PV and Dgen with PH245 storage)

In case 4, 1150 PV, 20 Dgen and 98 units of PH245 storage are required to meet the annual energy consumption of the RO system in Suez location. Which consider the most feasible system in this case with (NPC) \$13,600,475, and the (COE) is \$0.1200/kWh with a CEI of 132,753.03 kg/yr and an RF of 92.12 %.

5.2. Comparative analysis between different feasible HRESs cases

Five feasible cases were implemented and discussed, with a specific emphasis on the impact of employing renewable energy systems. The simulation results included optimal results of various (KPC). To facilitate an alternative method of comparison, [Table 15](#) provides an overview of the rankings of the five scenarios based on different aspects, including (NPC), (COE), Renewable Fraction (RF), Cumulative System Reliability (CSR), System Efficiency (SE%), Payback Period (PbP), and Carbon Emission Index (CEI).

5.2.1. Economic evaluation

Based on the optimal values presented in Tables, 8, and 9, it is evident that Case 3 holds the top position among all cases. Case 3 (PH245 + WT + PV) is the most cost-effective case, securing the first position with the lowest values of NPC, COE, and PbP among the five cases. The average values of NPC, COE, and PbP across all fourth locations are \$10,500,046, \$0.093/kWh, and 1.02 years, respectively.

Case 4 (WT + DGen + PH245) is ranked second, with NPC, COE, and PbP values of \$14,352,861, \$0.131/kWh, and 1.11 years, respectively. Additionally, the PH245 + WT (Case 2) is the third feasible case, with an NPC of \$14,720,240 and COE of \$0.130/kWh.

On the other hand, the PH245 + PV + DGen system (Case 1) occupies the fourth position, with NPC, COE, and PbP values of \$15,301,204, \$0.135/kWh, and 1.18 years, respectively. Case 5 is characterized by relatively high capital costs, resulting in an NPC, COE, and PbP of \$15,541,340, \$0.136/kWh, and 1.2 years, respectively.

These rankings provide a comprehensive assessment of the different configurations based on their cost-effectiveness, considering factors such as NPC, COE, and PbP.

5.2.2. Energy evaluation

Three factors can be used to assess any system's energy performance: (CSR), electricity production, and excess electricity. Referring to [Tables 12–14](#) and [Figs. 10–12](#) it can be observed that the PH245 + WT + PV system (Case 3) demonstrates the best economic performance. However, it has the lowest average electric production of 14,559,472 kWh/year and the smallest amount of excess electricity at 4,581,984 kWh/year. Consequently, Case 3 may not be beneficial and might not adequately meet future expansions.

Furthermore, with an CSR and UL 3650.87 kWh/year (0.107 %) and 4,581,984 kWh/year (0.07463 %) for the respective cases, this system exhibits remarkably low values for both CSR and UL. Despite these low percentages, [Table 6](#) demonstrates that Case 3 ranks third in terms of CSR."

Table 13
Total electric production of different scenarios.

	PV (Stand-alone) kWh/ year	WT(stand-alone) kWh/ year	Hybrid system (PV,WIND) kWh/year	Hybrid (wind, diesel) kWh/ year	Hybrid (PV, diesel) kWh/ year
Suez	15,382,088.0	13,855,499.0	13,983,298.0	13,432,332.0	14,637,613.0
Alex	13,791,753.0	25,983,981.0	13,387,233.0	24,007,564.0	14,856,475.0
Sallum	18,128,653.0	13,535,111.0	19,084,069.0	13,159,761.0	18,070,651.0
Shalteen	28,491,432.0	19,518,889.0	13,636,068.0	15,661,054.0	28,491,833.0
Average	16,805,202.67	17,645,284.33	14,559,472.00	15,895,997.00	17,338,731.50

Table 14

Excess electricity of different scenarios.

	PV (Stand-alone) kWh/ year	WT(stand-alone) kWh/ year	Hybrid system (PV,WIND) kWh/year	Hybrid (wind, diesel) kWh/ year	Hybrid (PV, diesel) kWh/ year
Suez	4,843,318.0	3,798,802.0	4,068,772.0	3,402,732.0	4,092,351.0
Alex	5,255,025.0	16,135,029.0	3,357,582.0	14,122,724.0	4,311,154.0
Sallum	7,629,640.0	3,502,231.0	9,370,522.0	3,113,777.0	7,571,611.0
Shalteen	18,009,523.0	9,461,221.0	3,558,209.0	5,503,416.0	18,009,823.0
Average	6,619,892.67	7,644,868.50	4,581,984.50	5,804,413.50	6,880,397.17

Table 15

Various performance aspects are used to calculate different rankings of the viable system.

Config	PV (Stand-alone)	WT(stand-alone)	Hybrid system (PV,WIND)	Hybrid (wind, diesel)	Hybrid (pv,diesel)
NPC	4	3	1	2	5
COE	4	2	1	3	5
PbP	4	2	1	3	5
REF	1	1	1	4	5
SE	3	2	1	4	5
CSR	5	4	3	1	2
CEI	1	1	1	4	5
Excess Elect	3	5	1	2	4

Conversely, the PH245 + DGen + PV system (Case 5) produces the highest amount of electric production at 17,338,731 kWh/year. This indicates that this configuration can effectively handle unexpected load fluctuations and future increases in plant load demand without significantly increasing system costs. Additionally, this system ranks fourth and second in terms of excess electricity and CSR, respectively, due to its ability to satisfy load requirements.

Furthermore, the technical analysis reveals that the PH245 + WT + DGen and PH245 + PV + DGen systems collectively rank first in CSR, as they meet the required electricity without any shortage.

Table 15 displays the ranking of all scenarios based on SE, CSR, and Excess Electricity.

5.2.3. Ecological evaluation

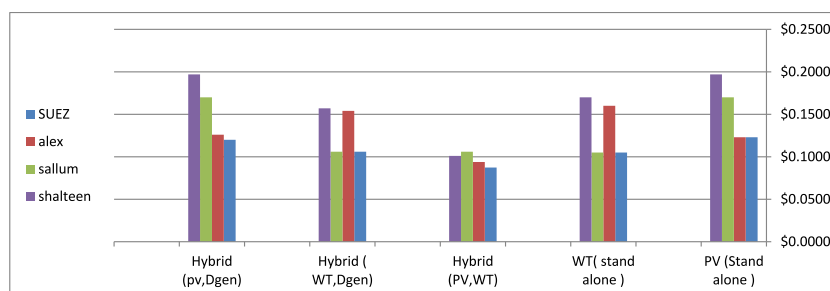
In this research, an environmental study is conducted by evaluating the (CEI), which is inversely proportional to the (RF), for the five optimal cases. Case 1, Case 2, and Case 3, which are pure renewable energy designs without the Diesel Generator (DGen), exhibit the highest RF due to zero emissions. Consequently, these systems rank first in both CEI and RF among all configurations.

Examining the results, it can be observed that the PH245 + Wind + DGen system (Case 3) secures the second position in terms of ecological behavior, with a 92 % RF. This configuration produces a realistic amount of carbon emissions, with only 102,536.22 kg/yr. Compared to the base-case of a diesel system, this results in a reduction of CEI by 96.22 %.

Furthermore, the emission results demonstrate that the PH245 + PV + DGen system is closely positioned in third place among all systems, with a CEI of 103,682.08 kg/yr and an RF of 91.12 %. This configuration achieves a reduction of 95.09 % in CEI compared to the base-case.

These findings highlight the positive environmental impact of the renewable energy-based configurations (Cases 1, 2, and 3) and the significant reductions in carbon emissions achieved when compared to the conventional diesel system.

To summarize, based on the previous evaluations and comparisons, no single system emerges as superior in all aspects of comparison. The PH245 + Wind + PV system (Case 3) demonstrates the best economic performance, but it falls short in terms of energy reliability. On the other hand, the PH245 + Wind + DGen system (Case 4) excels in load feeding reliability but ranks lower in economic terms due to its high capital cost. Cases 1, 2, and 3 achieve 100 % Renewable Fraction (RF) and have no Carbon Emission Index (CEI), making them sustainable and ecologically favorable. However, from an economic standpoint, they may not be the best choices. It is

**Fig. 8.** Cost Analysis of optimal systems (cost of electricity).

evident that different performance aspects influence the ranking of the systems. Therefore, Multi-Criteria Decision Making (MCDM) is employed, utilizing Fuzzy-AHP to select the optimal configuration. This analysis highlights the importance of considering multiple criteria and trade-offs when evaluating and selecting the most suitable system, as each configuration has its strengths and weaknesses in different performance aspects.

5.3. Cost analysis of the optimal system

Figs. 8 and 9 show the cost analysis and net present cost of the different scenarios [standalone PV, standalone wind farm, hybrid system (PV and diesel), hybrid system (Wind and diesel) and hybrid system (PV and Wind)]. By comparing the results, we see that the hybrid system has lower values of COE and NPC compared to the stand-alone PV system. For the hybrid system, the highest cost of energy (COE) obtained is 0.1060\$/kWh for Sallum while Alexandria, Shalteen and Suez have 0.0938, 0.1010, and 0.0873 \$/kWh as their COE respectively. The lowest COE obtained is 0.0873\$/kWh for Suez. The highest net present cost (NPC) is 12 million \$ for Sallum, while Alexandria, Suez, and Shalteen have 10.6, 9.89, 12 million \$ respectively. The lowest (NPC) is 9.9 million for Suez.

5.3.1. Optimization of system parameters

Tables 8, 9 and 10 give the optimized system parameters according to electrical demand for each location.

5.4. Sensitivity analysis

A sensitivity analysis was conducted on a hybrid photovoltaic (PV), wind, and storage PH245-powered reverse osmosis (RO) desalination system. This analysis aimed to assess the system's sensitivity to changes in various parameters and identify their impact on system performance and economics.

The parameters considered for sensitivity analysis included.

- Location: The variation in location would result variation in renewable resources. The impact of site location is implemented at four locations
- The effect of plant load growth is considered at constant renewable energy resources for Suez location. The load growth was assumed to be 32 % and 62 % of the specified value.
- The effect of interest rate mainly depends on the economic status of the state; two scenarios were implemented once by increasing the discount rate to 10 % and once again by reducing its value to 5 %.
- Storage capacity: The storage capacity was varied to examine its influence on the system's ability to store excess energy and provide a stable water supply during periods of low generation. The impact of reducing the storage cost by 50 % and 70 % is examined.
- Loss of power supply probability (LPSP): The LPSP was varied to examine the fraction of wind turbine fraction.

Figs. 13 and 14 shows line plots of location variable based on NPC and COE. For Fig. 13 (a), variation in location leads to solar and wind speed value varies. The analysis reveals that increasing the wind speed and solar radiation by 8 % and 11 % will decrease the NPC and COE by 21 %.

The results show that higher wind speeds result in increased wind power generation, complementing the PV system's output.

Figs. 15 and 16 shows the response of the NPC and COE against the variations in the three considered sensitivity variables. Specifically, the analysis reveals that a 20 % increase in load demand leads to a 21.2 % rise in the NPC, while a 40 % increase in load demand corresponds to a 39.2 % increase in the NPC when using the TPH energy storage. This increase of system cost is due to the substantial growth of the direct investment for the further PV panels, WTs, and energy storage units to satisfy the load growth and enhance the system reliability.

In the view of the ESTs cost, reducing the cost of the TPH storage by 50 % and 70 % executes lessening in both NPC and COE by 11.7

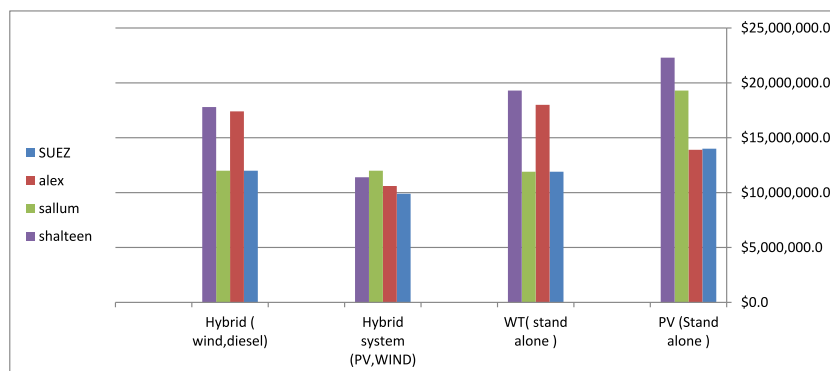


Fig. 9. Cost Analysis of optimal systems (Net Present cost).

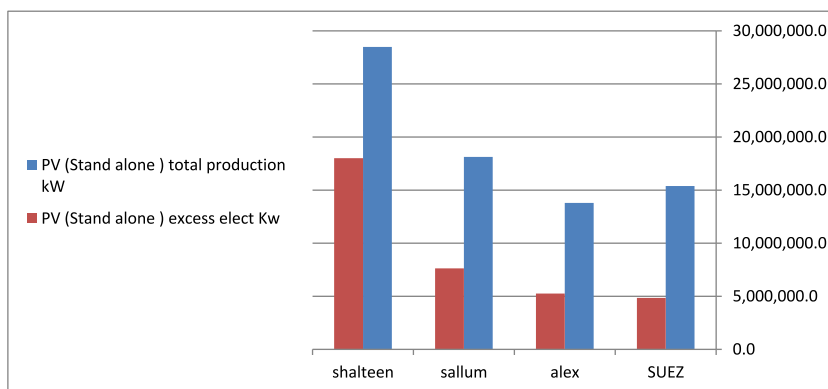


Fig. 10. Total production & excess electricity (PV stand-alone).

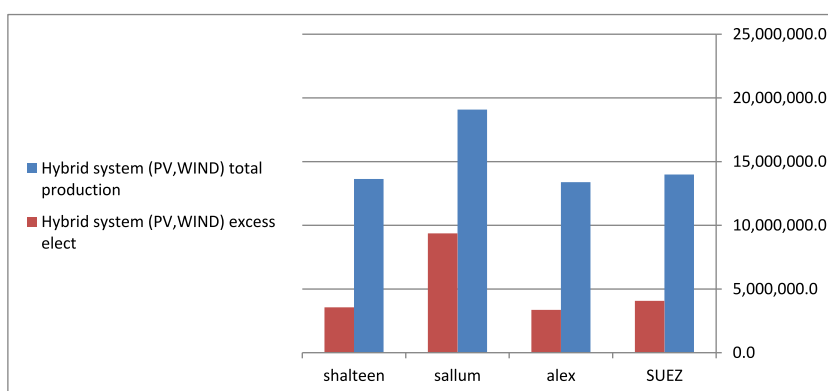


Fig. 11. Total production & excess electricity hybrid system (PV,Wind).

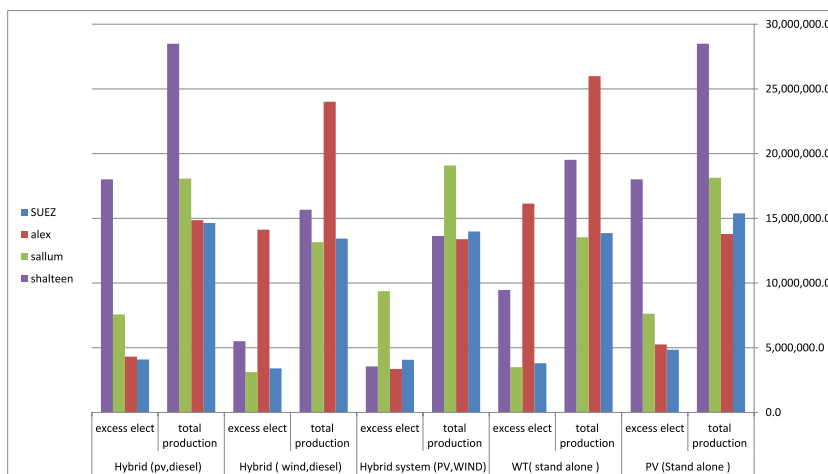


Fig. 12. Total production & excess electricity for different scenario.

% and 6.57 %, respectively.

Upon further analysis, it becomes evident that the interest rate has a noteworthy influence on the Net Present Cost (NPC) and the Cost of Energy (COE). The results indicate an inverse relationship between the interest rate and the NPC, while a direct relationship exists between the interest rate and the COE. Consequently, a potential decrease in the interest rate to 3 % leads to a substantial 8.2 % increase in the NPC when utilizing the PH245 energy storage system. Simultaneously, it results in a significant reduction in the COE to \$0.0821/kWh. Conversely, if the interest rate were to rise to 12 %, the NPC would decrease by 8.7 %, but the COE would increase to

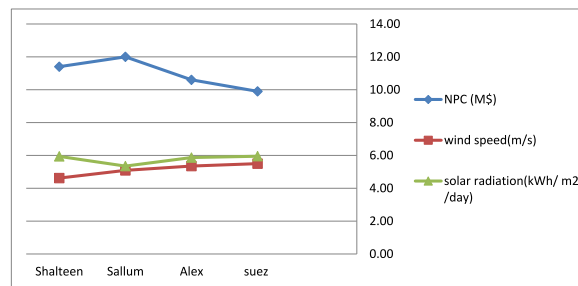


Fig. 13. Location variable based on NPC.

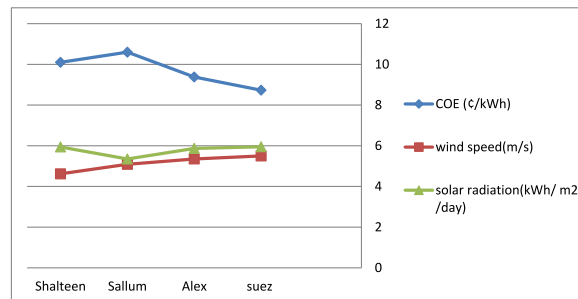


Fig. 14. Location variable based on COE.

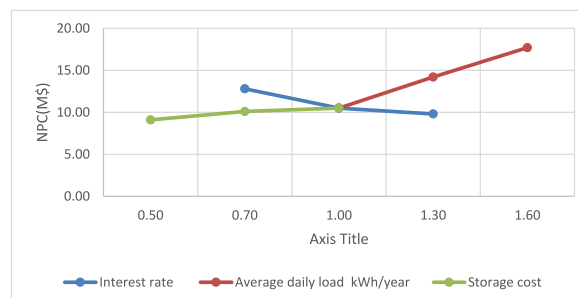


Fig. 15. Sensitivity variables based on NPC.

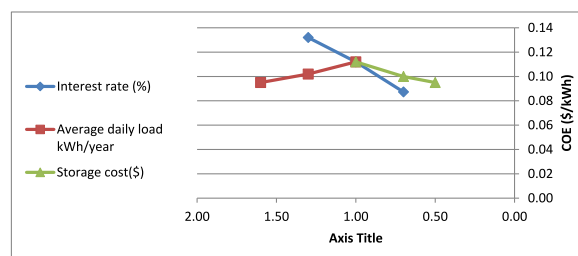


Fig. 16. Sensitivity variables based on COE.

\$0.135/kWh.

At Suez location Fig. 17 shows the loss of power supply probability (LPSP) increases, the fraction of wind turbine production increases. Therefore, the optimal configuration requests more wind turbine. For the LPSP equals 0 % and 2 % the minimum percentage of wind turbine fraction is 43 % and 71 % respectively. However, as regards LPSP equals 10 %, the fraction of wind turbine production is 100 %.

These findings highlight the importance of carefully considering site selection and renewable energy resources in the selected

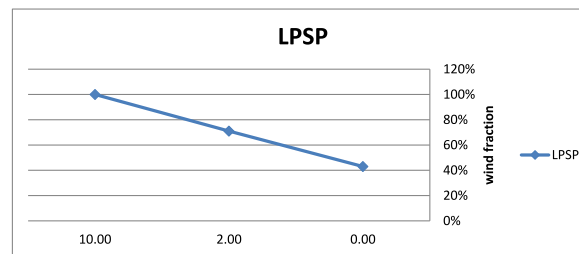


Fig. 17. LPSP variable based on wind fraction.

optimal system when making investment decisions. Decision-makers need to take these factors into account to make informed choices that align with the specific requirements and objectives of the project.

5.5. The analysis of the socioeconomic impacts of (HRESs) powered (RO)

Using Reverse Osmosis (RO) desalination plants powered by Hybrid Renewable Energy Systems (HRES) offers several advantages that can positively impact socioeconomic factors.

Here are some key advantages:

Cost Savings: HRES-powered RO desalination plants can lead to significant cost savings compared to fossil fuel-powered plants. Renewable energy sources, such as solar and wind, have lower operating costs once the initial investment is made. This can result in reduced water production costs, making desalinated water more affordable and accessible to communities, businesses, and industries.

Job Creation and Local Economy: The deployment of HRES-powered RO plants can create new employment opportunities and stimulate the local economy. The development, installation, operation, and maintenance of renewable energy systems and desalination plants require skilled labor. This can lead to job creation and economic growth, benefiting local communities and contributing to sustainable development.

Energy Independence and Security: HRES-powered RO plants reduce dependence on imported fossil fuels, improving energy security and independence. By harnessing renewable energy sources locally, communities and countries can reduce their vulnerability to fuel price fluctuations and supply disruptions, ensuring a stable and reliable energy supply.

Environmental Sustainability: HRES-powered RO systems have a significantly lower carbon footprint compared to fossil fuel-powered plants. By relying on clean and renewable energy sources, such as solar and wind, the greenhouse gas emissions and environmental impacts associated with energy generation are significantly reduced. This contributes to mitigating climate change, preserving natural resources, and improving overall environmental sustainability.

Water Availability and Security: HRES-powered RO desalination plants can enhance water availability and security, particularly in regions with limited freshwater resources. By utilizing renewable energy sources, these plants can operate reliably and continuously, ensuring a consistent supply of clean water. This is crucial for supporting various sectors, such as agriculture, industry, and domestic usage, and improving overall water security.

Community Resilience and Empowerment: HRES-powered RO systems can enhance community resilience by providing decentralized and sustainable water solutions. These systems can be deployed in remote or underserved areas, reducing dependency on external water sources and empowering communities to meet their water needs independently. This enhances community resilience and contributes to social and economic development.

By leveraging the advantages of HRES-powered RO desalination plants, communities can benefit from improved water affordability, job creation, energy independence, environmental sustainability, and overall socioeconomic development.

Table 16

Calculated results compared with most recent researches.

reference	year	location	COE
[22]	2020	Baltim, Egypt	0.1019 \$/kWh
[54]	2023	west coast of Türkiye	0.529 \$/kWh
[55]	2023	Mexico	0.480 \$/KWh
[56]	2020	Algeria	0.210 \$/KWh
[57]	2022	Saudi arabia	0.0557 \$/kWh
[58]	2020	China	0.2345\$/kWh
[59]	2022	India	0.140\$/kWh
[60]	2023	Iran	0.1325 \$/kWh
Current research	2023	Suez, Egypt	0.0873 \$/kWh
		Alex, Egypt	0.0938 \$/kWh
		Sallum, Egypt	0.106 \$/kWh
		Shalteen, Egypt	0.101 \$/kWh

5.6. Evaluation of model accuracy compared with relative literature

However, comparing the analysis of this study with related literature proved challenging due to variations in the structure of the Hybrid Renewable and Storage Energy System (HRES), renewable resources, and load demand. However, economic measures served as a sufficient benchmark for assess and the obtained outcomes with those reported in the literature. Table 16 provides a concise evaluation, comparing the Cost of Energy (COE) values from this study with absolute values from specific HRESs projects implemented in various locations worldwide.

As the Net Present Cost (NPC) calculation depends on component capacities and project capital expenses, its significance varied significantly from one region to another, making direct comparisons difficult. "On the other hand, COE values can be considered a reliable indicator of the cost of sustainable development. An analysis of Table 16 reveals that the case study conducted in Turkey [54] had the highest COE value of 0.529 \$/kWh, whereas the case study mentioned in Ref. [57] exhibited the lowest COE (0.0557 \$/kWh) among the other locations.

The average cost of energy (COE) for the electricity in several case studies was determined to be 0.23545 \$/kWh. However, the COE for the current study was found to be lower than this average. This finding supports the economic viability of the proposed HRES in the selected areas and demonstrates a significant alignment with previous international investigations."

5.7. Pros and cons of the proposed hybrid system configuration

Integrating hybrid renewable energy into desalination systems in Egypt has several advantages and disadvantages. Here are some pros and cons.

5.7.1. Pros

Energy Independence: Egypt can reduce its reliance on fossil fuels and achieve energy independence. This lead to greater energy security and reduced exposure to volatile fuel prices.

Cost Savings: proposed system has the potential to provide cost savings in the long run. Once the initial investment is made, the operational costs are relatively low compared to conventional energy sources, such as diesel generators.

Sustainable Water Production: the proposed system provides a sustainable solution to meet Egypt's growing water demand.

Environmental Sustainability: The integration of renewable energy into desalination systems reduce greenhouse gas emissions and mitigate climate change. It contributes to Egypt's efforts in meeting its carbon reduction targets and promoting a cleaner and greener environment.

Water-Food-Energy Nexus: Integrating renewable energy with desalination systems can address the interconnection between water, food, and energy security. It provides a sustainable solution to meet the increasing demand for freshwater, which is essential for agriculture, food production, and overall economic development.

Local Job Creation and Economic Development: The development and deployment of renewable energy technologies can create job opportunities and stimulate economic growth in Egypt. It can attract investments in the renewable energy sector, promote research and development, and foster a skilled workforce.

5.7.2. Cons

Initial Investment Cost: The upfront investment required for establishing hybrid renewable energy systems can be significant.

Intermittency and Variability: Renewable energy sources such as solar and wind power are intermittent and subject to natural variability. This can lead to fluctuations in energy generation, which impact the stability and reliability of desalination systems. Adequate energy storage or backup systems are necessary to ensure a continuous and reliable water supply.

Maintenance and Operation: Ongoing maintenance and operation of hybrid renewable energy systems require skilled technicians and regular upkeep. Proper maintenance is crucial to ensure the longevity and efficiency of the systems, which may involve additional costs.

6. Conclusion

- The aim of this study is to conduct a comprehensive feasibility analysis of a hybrid PV/wind system for powering a desalination unit in various locations in Egypt, including the Mediterranean seacoast and Red Sea coast. These regions are known for their abundant PV and wind energy potential. The study also involves comparing the proposed hybrid system with other alternative systems using HOMER software to determine the most suitable option. The research focuses on optimizing the design and evaluating the Hybrid Renewable and Storage Energy (HRSE) system, which consists of PV panels and wind turbines.
- Five different system options are considered: stand-alone PV, stand-alone wind farm, hybrid system with PV and diesel backup, hybrid system with wind and diesel backup, and hybrid system with both PV and wind.
- The load demand in this study represents a large-scale reverse osmosis (RO) plant that supplies freshwater to remote areas in Egypt. The plant has a rated capacity of 15,000 cubic meters per day, with an average demand of 1833 kWh and a peak demand of 3000 kW.
- By analyzing these parameters and using the HOMER software, the study aims to determine the optimal configuration and evaluate the feasibility of the proposed hybrid PV/wind system for meeting the water demand in these specific locations. The optimization

based on finding the minimum net present cost (NPC) feeds the load demand and maintains the system's reliability. Secondly, the (MCDM FAHP) is implemented to select the optimal system considering ten performance criteria

The significant results of the investigations can be summarized as follows.

1. Optimization results indicate that the WT + PV + PH245 system (Case-3) is the most cost-effective option, exhibiting the lowest Net Present Cost (NPC), Cost of Energy (COE), and Payback Period (PbP) values of \$10,523,121, 0.093 \$/kWhr, and 1.02 years, respectively. However, this configuration has the lowest average electric production of 14,559,472 kWh/year and the smallest amount of excess electricity generated, which stands at 4,581,984 kWh/year. Case-3, the WT + PV + PH245 system, demonstrates the best ecological impact among all the alternatives considered in the study.
2. The utilization of the PH245 storage system instead of batteries makes the WT + PV + TPH configuration the most favorable system for supplying the Reverse Osmosis Desalination Plant (RODP).
3. In conclusion, the investigation shows that the WT + PV + PH245 system (Case-3) offers the most cost-effective solution with a favorable ecological impact. However, it should be noted that this configuration has slightly lower electricity production and excess electricity compared to other alternatives. On the other hand, the PH245 + DGen + PV system (Case 5) demonstrates the highest electricity production, generating 17,338,731 kWh/year. This indicates that this particular configuration is well-equipped to handle unexpected load fluctuations and future increases in plant load demand without a significant increase in system costs.
4. The results of the (MCDM) Fuzzy-AHP analysis reveal that Case 3, which consists of 7×20 -kW wind turbines, a 4909-kW photovoltaic array, 152 PH245 storages, and a 737-kW converter, provides a suitable trade-off solution for supplying the (RO) facilities and meeting the freshwater needs of the rural areas. This configuration is particularly advantageous due to the favorable solar radiation conditions in selected site. Similarly, in the Sallum site, Case 3, which includes 38×20 -kW wind turbines, a 2938-kW photovoltaic array, 184 PH245 storages, and a 510-kW converter, is recommended. This is due to the advantageous wind speed conditions in Sallum, making it an optimal choice for meeting the energy demands of the RO facilities in that location.
5. In the case of a stand-alone PV system, the cost of electricity production across the studied sites is found to be 0.123, 0.123, 0.17, 0.197 \$/kWh, respectively. However, in the case of the hybrid system, the cost of electricity production decreases to 0.0873, 0.0938, 0.106, 0.101 \$/kWh across the same sites. This clearly demonstrates that the PV/wind combination in the hybrid system is more cost-effective due to the complementary nature of the two energy sources.

The sensitivity analysis indicates that:

1. The hybrid combination of PV and wind energy is more economically and technically viable compared to a stand-alone PV system. The total net present cost of the hybrid system is lower than that of the stand-alone PV system in different locations in Egypt.
2. By utilizing a hybrid system, the number of batteries and converters required in the system can be reduced, resulting in a lower total net present cost. This not only enhances the reliability of the system but also ensures that it can meet the load demand effectively. Consequently, this research highlights the feasibility, reliability, and cost-effectiveness of installing PV/wind hybrid systems to provide electricity to remote areas throughout Egypt, compared to stand-alone PV systems.
3. Using of fresh water storages instead of batteries improves system reliability and reduce capital and maintenance costs. This is due to the relatively short lifetime of battery systems. By adopting this approach, the hybrid system becomes more dependable and cost-efficient, making it a favorable choice for remote area electrification in Egypt.

6.1. Future work

In the future work we can delve into the weaknesses of the proposed method and suggest potential avenues for improvement. Here are some possible directions they could explore.

- **Performance Validation:** The authors can consider conducting performance validation of the proposed method by comparing the predicted results with real-world data from existing hybrid PV/wind powered desalination systems in Egypt or similar locations. This would help assess the accuracy and reliability of the method in capturing the actual performance of such systems.
- **Optimal Control and Dispatch Strategies:** The study can explore advanced control and dispatch strategies for the hybrid PV/wind powered desalination system. This could include optimizing the power allocation between PV and wind sources based on real-time solar/wind data, load demand, and energy storage status. The authors can consider advanced algorithms, such as model predictive control or reinforcement learning, to optimize system performance and maximize energy utilization.
- **Scalability and Integration:** we explore the scalability and integration potential of the proposed hybrid system with existing infrastructure. This includes evaluating the compatibility of the system with the grid, assessing the feasibility of integrating the system into existing desalination plants or water distribution networks, and considering the potential for hybrid systems at larger scales.
- **Integration of Water Demand Variability:** We can consider incorporating the variability in water demand into their analysis. Water demand patterns may fluctuate seasonally or based on specific events, and accounting for these variations can provide more accurate assessments of system feasibility and sizing. This could involve analyzing historical water consumption data, considering future population growth projections, or incorporating factors like tourism or agricultural water needs.

- **Reliability and Resilience Analysis:** The study can delve into the reliability and resilience of the hybrid system by considering factors such as equipment failure rates, maintenance downtime, and the impact of extreme weather events. By incorporating these aspects, the authors can assess the system's ability to consistently provide reliable water supply and identify strategies to enhance its resilience, such as redundant components or backup power sources.

CRediT authorship contribution statement

Ibrahim Elsayed: Project administration, Methodology, Conceptualization. **Hamdy Kanaan:** Writing – original draft, Software, Conceptualization. **Mohammed Mehanna:** Writing – review & editing.

Data availability statement

All data included in this study are available.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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